

Global Facilities - Design & Construction Standard – Electrical - On-Site Blackout Test

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Owner(s)	ENGINEERING	Target Audience:	All Facilities
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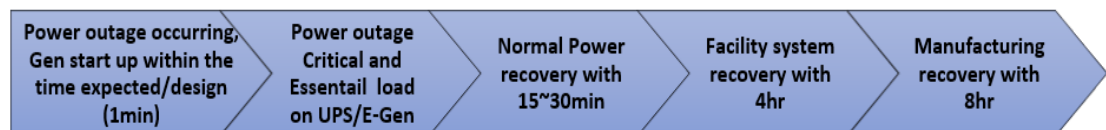
[ECN History](#)

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1.0 Purpose

- 1.1 Safety First! To ensure the safety of people in the environment when a blackout.
- 1.2 Conducting a blackout test is a must do test as part of commissioning critical facilities before new building into operation/manufacturing, such as manufacturing(FAB), operations centers, and essential facilities infrastructure.
- 1.3 The goal of the blackout testing process is to verify full, integrated system functionality that maintains safety for the users.
- 1.4 The blackout test is an opportunity to uncover potential safety oversights, even in fully functional systems, before they become an issue. Operations staff gain invaluable training through their participation in these scenarios.
- 1.5 A successful blackout test yields peace of mind for facility owners and operations staff, proving that systems work, under the worst-case conditions, and are ready for action when emergencies happen.

Flow chart of the blackout test stage as following for reference:



2.0 Safety

2.1 Safety Documents

Compliance with all Corporate Safety Policies local site safety policies and local code or regulatory needs as required.

2.2 Personal Protective Equipment (PPE)

The switching procedure will identify the PPE required for the work.

2.3 Hazard Identification

Electrical shock and arc-flash hazards are present during switching operations. Any other potential hazards shall be identified and precautionary measures to be taken.

3.0 Scope

3.1 The document covers the new Base build System installations only and does NOT cover tool installation.

3.2 Site (s) Impacted:

This standard specifies requirements and provides recommendations for the design and construction of fixed electrical installation in Micron. It applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

4.0 Definitions and Acronyms

Term/Acronym	Definition/Description
The Contractor	The party that carries out all or part of the engineering design, purchasing, installation, testing and commissioning of a project. The Owner may undertake all or part of the duties of the Contractor.
The Manufacturer/Supplier	The party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.
The Owner	The party that initiates the project and eventually pays for it. The Owner may also include an agent or consultant authorized to act for, and on behalf of, the Owner.
High Voltage (HV), 601-01-27, IEC 60050	The set of voltage levels in excess of low voltage. Although not covered by an IEC definition, it has become industry practice to use the term Medium Voltage (MV) for voltages from 1 kV up to 35 kV.
Low Voltage (LV), 601-01-26, IEC 60050	A set of voltage levels used for the distribution of electricity and whose upper limit is 1000 V AC
Extra Low Voltage (ELV)	Electrical systems that operate under 120 V DC (ripple free) or 50 V AC. ELV systems do not require an electrical license to be installed or worked on.
Blackout Test	The Power Supplies to the Operating Facility are deliberately Failed to confirm Emergency and D/S/ UPS Supplies 'start up' and maintain essential systems
Switching Procedure	The appropriate Facilities personnel shall write the switching procedure before any work is performed.
FMCS	Facilities Management and Control System
E-Gen	Emergency Generator system.
DUPS	Dynamic Un-interruption Power Supply
SUPS	Static Un-interruption Power Supply
ATS	ATS Automatic Transfer Switch
SMS	Sag Mitigation System
GFTT	Micron Global Facility Technology Team
FCT	Facilities Central Team.
EHS	Environmental, Health, Safety.
CoHE/LOTO	Control of Hazardous Energy/ies or sometimes called Lockout/Tagout. Preventing machinery/equipment from

Term/Acronym	Definition/Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

5.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

6.0 Blackout Test

6.1 Objectives:

- Ensure the facilities systems interdependencies respond as specified in the design criteria by testing the integrated systems in the following conditions.
- Abnormal Operations Condition (Power Interruption)
- Test failure modes and system restart.
- Ensure reliability of site critical systems.
- Verify system capacities.
- Provide a valuable training exercise for the following groups: ERT, Security, sustaining operations and engineering staff.
- Facilities team needs to provide a test outcome document to list the systems reaction.

6.2 Prerequisite

The Facilities engineering group owns the development of the blackout procedures(A switching procedure developed by Electrical and Mechanical) and identification of resource requirements for the execution of the tests. The blackout owner for each test develops a detailed Blackout test Gantt chart outlining each task, with resource requirements and recovery plan. The Facilities Engineer ensures the blackout team follows the SIPP Process during the testing. The blackout must be built into the Project Master Schedule. Blackout should occur after critical systems are completed and coordinated with Tool Install processes. Two planned dates should be considered to allow for addressing any major issues and be able to retest.

6.3 Staffing

One Facilities Senior Electrical and one Facilities Senior Mechanical Engineer share the leadership role. Facilities Senior Technicians share development and execution of blackout tests with the Facilities Engineering Group. Support should be broken down by system and include representation from Facilities and Construction Teams.

6.4 Roles and Responsibilities:

- Facilities Engineering leads the blackout process.
- Existing System Teams support the blackout of their specific systems.
- Facilities and Construction Teams support the blackout planning effort. The Projects is responsible for the funding requirements associated with blackout including providing contractor resources for the set-up and execution of the tests. The Project Engineers own review and approval of the blackout procedure.
- Fac Engineering is responsible for ensuring the tests and durations are incorporated into the Project Master Schedule.
- Facilities Engineering Manager owns the communication link to Facilities Management and obtaining support for the blackout process.
(Project Funding would need to be allowed for. at Project outset or additional Capital will be required if needed.)
- The Construction team and Commissioning team are responsible for develop the switching procedure.

6.5 Blackout activities

Blackout is considered complete when the systems pass the required tests and are ready to operate in a continuous mode.

Examples of typical tests:

- Emergency Power System Test
- Diesel/Gas Power Generating/DDUPS/SUPS System Test
- Normal Power System Test
- Emergency Power System Load Test
- Make-up air controls and system operations test
- Smoke exhaust test to validate smoke removal through the use of tracer gas.
- Fire suppression systems/fire shutters, etc. test.
- Instrument air test to determine system impacts during loss of instrument air.
- Process cooling water test-to-test system response during a simulated line break.
- Ultra Pure Water System Test
- Acid Waste Neutralization System Test
- Facility Management Control System / Power SCADA Test
- IT systems redundancy and loading/capacity test.
- Acid, General, Ammonia and VOC Exhaust systems test.
- ...etc.

7.0 Conducting Test activities:

Ready,

Preparation for the blackout test begins by assembling the team. A scripted flow of test sequences and verification steps for each person is outlined. Personnel are distributed throughout the facility/infrastructure, with a test script in hand, ready to test the system. They will observe and record actions and responses according to each role.

Ensure all the systems are operating per the testing procedure and been commissioning per appendix list, refer to appendix A-Blackout test check list.

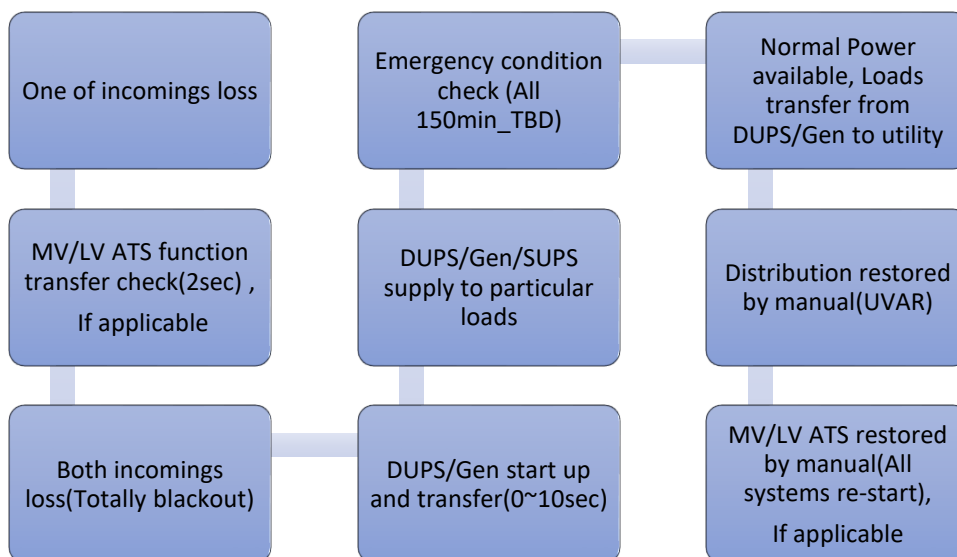
Set-up,

Once in position, each team member verifies that systems are in normal operation: no maintenance is in progress that would impede testing, onsite crews are aware and notified prior to initiation of the blackout, the generator has sufficient fuel to endure the test, lighting is on, confined spaces are certified safe to access, systems are not in test mode, local authorities(Including Utility company) have been notified, user/password credentials are in-hand, monitoring and protection systems are active, and equipment and elevators are operating normally. We haven't even started the test yet! Coordinating a team around this level of integration requires a high level of preparation, communication, and patience.

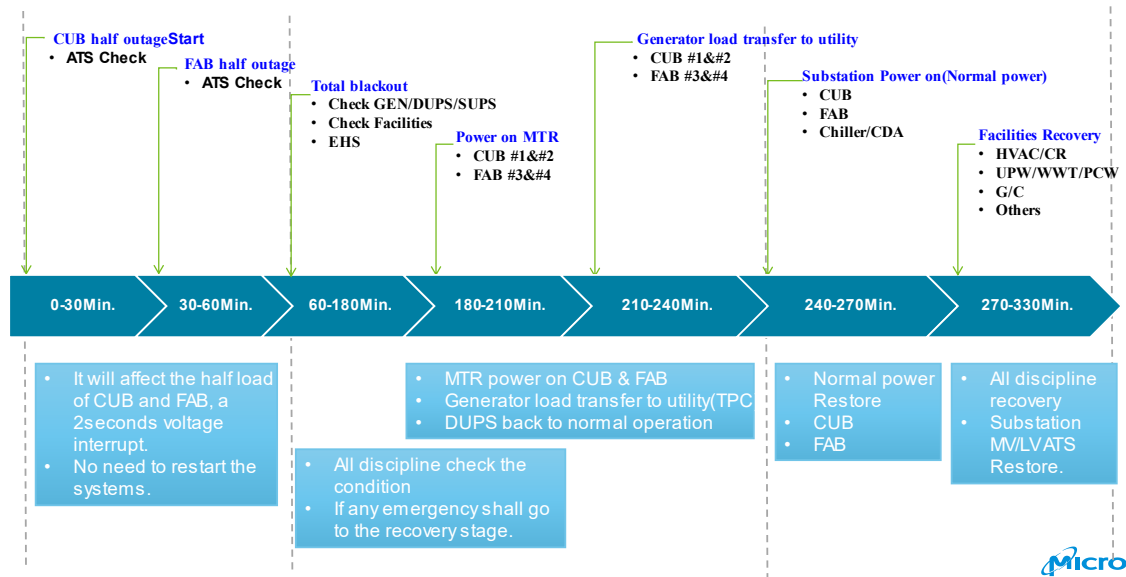
GO!

After verifying normal operation, the team is ready to initiate the blackout test. Everything from here on out is triggering/simulating conditions and observing system responses.

Blackout Testing Flowchart example:



Time chart example based on F16 testing:



System Restore,

The blackout test concludes by returning all systems to normal operation, such as powering down the backup generator and switching back to normal utility power. All systems are verified to return to normal operation on utility power, restored to normal condition, and backchecked against the conditions recorded before the blackout test.

Systems were operated normally and monitored for any errors for a month after the blackout/stress test. At this level of infrastructure, redundant safety checks are part of the plan, and all aspects of the process are verified and reverified to establish a deep level of confidence before the infrastructure is opened/released to the manufacturing.

8.0 Lesson learnt and Close out.

Assign Owners to the list of issues collected during the testing and documented, track on the open issues until closed out all issues.

Prepare for a second test if first one was not carried out successfully.

9.0 Document Control

9.1 **Approval:** GLOBAL_FAC_GFTT_APPR

9.2 Notification:

GLOBAL_FAC_MANAGERS
 GLOBAL_FAC_COMMUNICATIONS
 GLOBAL_FAC_CONSTRUCTION
 FCG_F4_FAC_NOTIFY
 FCG_F6_FAC_NOTIFY
 FCG_F10_FAC_NOTIFY
 FCG_F11_FAC_NOTIFY
 FCG_F15_FAC_NOTIFY
 FCG_F16_FAC_NOTIFY
 FCG_MMP_FAC_NOTIFY
 FCG_MMY_FAC_NOTIFY
 FCG_MSB_FAC_NOTIFY
 FCG_MTB_FAC_NOTIFY
 FCG_MXA_FAC_NOTIFY

9.3 Final Viewer Access

All Micron Team Members at all Sites

9.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#)

9.5 Review

Biennially (two years)

10.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	02/15/22

11.0 APPENDIX

APPENDIX A: Blackout test check list

Blackout test preparation checklist before initial Testing example:

Discipline	Condition/Documents (Testing and commissioning)	Owner	Load list check Y/N	Commissioning Completed Y/N	Risk assessment Done Y/N	Check list ready Y/N	Contingency plan for recovery Y/N	Comment	Note
Electrical	Electrical system study								
Electrical	Normal Power System Test(Substation and Network)								
Electrical	Emergency Power System Test(Substation and Network)								
Electrical	Diesel Power Generator System Test								
Electrical	DUPS System Test								
Electrical	SUPS System Test								
Electrical	SMS System Test								
Electrical	Blackout testing Procedure(SIPP/PTP)								
Electrical	Electrical Check List(before/ active /after)								
Electrical	Power SCADA system test								
Electrical	FMCS test								
Electrical	Fire protection system test								
Mech.	Clean Room condition study								
Mech.	MAU controls and system operations test								
Mech.	Exhaust controls and system operations test								
Mech.	FFU controls and system test								
Mech.	Chiller water systems test								
Mech.	CDA system test								
Mech.	CR system Check List(before/ active /after)								
Mech.	HVAC system Check List(before/ active /after)								
Water	Ultra Pure Water System Test								
Water	Waste Water Treatment System Test								
Water	PCW system test								
Water	Ultra Pure Water System Check List(before/ active /after)								

Water	Waste Water Treatment System Test Check List(before/ active /after)								
Water	PCW system testCheck List(before/ active /after)								
Chemicals	Bulk Chemicals test								
Gas	Bulk + Specialty Gases test								
Chemicals	Bulk + Specialty Gases Check List(before/ active /after)								
Gas	Chemicals Check List(before/ active /after)								
EHS	EHS Check List(before/ active /after)-Safety								
EHS	EHS Check List(before/ active /after)-Environment								
EHS	EHS Safety/People Restriction plan								
EHS	CCTV system Check List								
	Facility Management System Test								
BCP	BCP follow chart								
CSA	Facilities System Test(Elevator...etc.)								
EHS	Provide a valuable training exercise for the following groups: ERT, Security, sustaining operations and engineering staff								
IT	IT CUP room and Tel check list								